CBSE Class 7 · Mathematics (Ganita Prakash, Part 2)

Geometric Twins — Homework

Congruence of figures and triangles · 30 questions

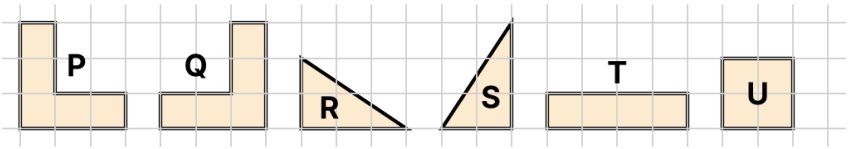
Name: _____ Roll No.: _____ Date: _____

Instructions: Answer all questions in your notebook. Give reasons for every step. Equal marks (|, ||) on sides show equal sides; equal arcs on angles show equal angles. Figures help you to see the situation — always use the measurements given.

Section A — Congruent Figures and Corresponding Parts

1. Are the figures in each pair **always** congruent? Write Yes or No and give a reason.
(a) Two line segments, each 5 cm long.
(b) Two squares, each of side 4 cm.
(c) Two circles, each of radius 3 cm.
(d) Two rectangles, each of area 12 cm².
(e) Two angles, each measuring 40°.
(f) Two equilateral triangles, each with perimeter 18 cm.

2. Shapes P, Q, R, S, T and U are drawn on a square grid.
(a) Which pairs of shapes are congruent?
(b) Shapes T and U have the same area. Are they congruent? Explain.



3. ΔABC ≅ ΔPQR. Write all **six** pairs of equal corresponding parts (three pairs of sides and three pairs of angles).

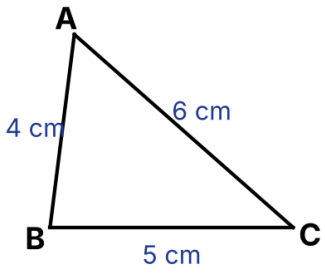
4. ΔXYZ ≅ ΔLMN. In ΔXYZ, XY = 6 cm, ∠X = 50° and ∠Y = 60°. Find:
(a) LM (b) ∠L (c) ∠M (d) ∠N

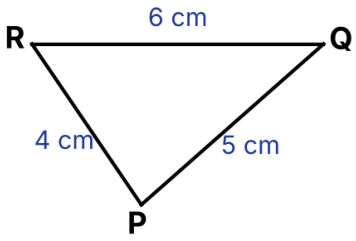
5. State True or False. Give a reason for each answer.
(a) Two triangles whose three angles are equal (one to one) are always congruent.
(b) If two figures are congruent, they have equal areas.
(c) If two figures have equal areas, they are congruent.
(d) Two right-angled triangles with equal hypotenuses are always congruent.
(e) Two triangles with equal perimeters are always congruent.
(f) Every figure is congruent to itself.

6. Look at the two triangles below.

(a) Are ΔABC and ΔPQR congruent? Which criterion tells you so?

(b) Riya wrote "ΔABC ≅ ΔPQR". Is she correct? Write the correct congruence statement.

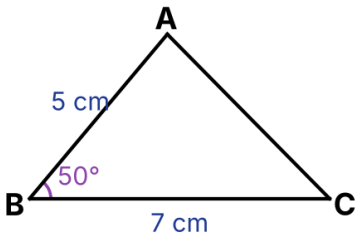


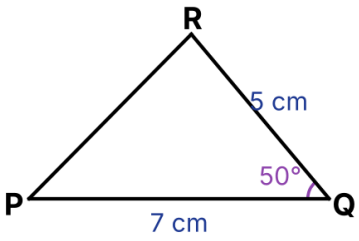


Section B — Criteria for Congruence (SSS, SAS, ASA, AAS, RHS)

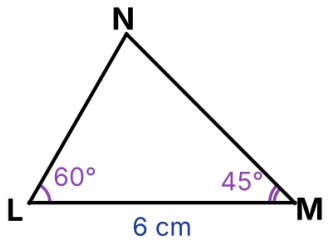
For Questions 7 to 10, look at the marked measurements. Decide whether the two triangles are congruent. If they are, name the criterion and write the congruence statement with the vertices in the correct order.

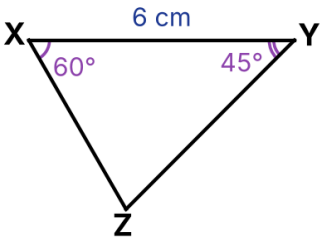
7. Are the two triangles congruent? If yes, name the criterion and write the congruence statement.



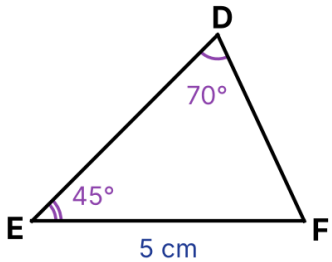


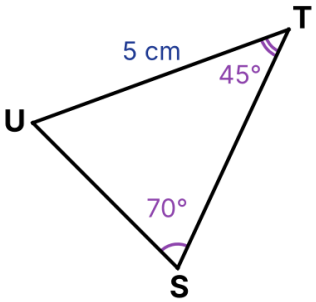
8. Are the two triangles congruent? If yes, name the criterion and write the congruence statement.



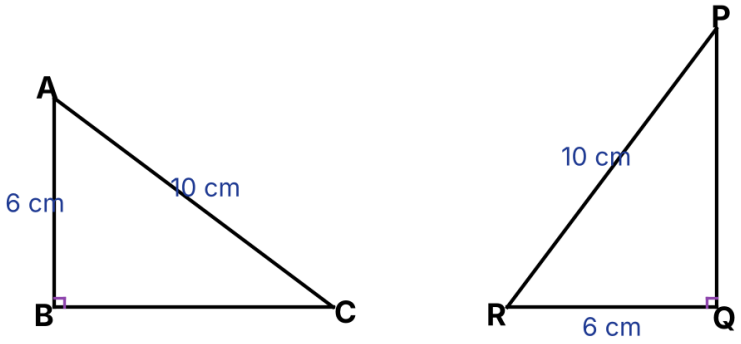


9. Are the two triangles congruent? If yes, name the criterion and write the congruence statement.





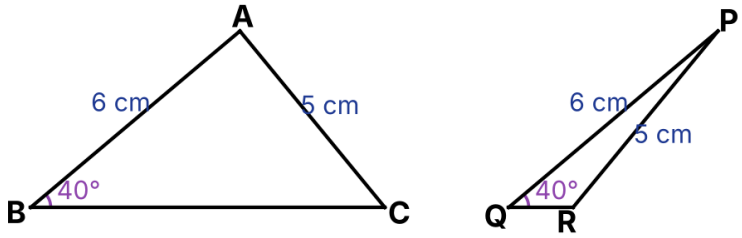
10. Are the two triangles congruent? If yes, name the criterion and write the congruence statement.



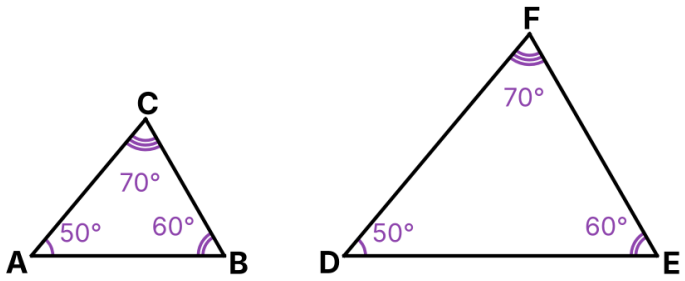
11. In the figure, AB = PQ = 6 cm, AC = PR = 5 cm and ∠B = ∠Q = 40°.

(a) Three parts of ΔABC are equal to three parts of ΔPQR. Are the two triangles congruent?

(b) Compare BC and QR in the figure. Explain why "two sides and a non-included angle" (SSA) is **not** a criterion for congruence.



12. In ΔABC and ΔDEF, ∠A = ∠D = 50°, ∠B = ∠E = 60° and ∠C = ∠F = 70°. Is ΔABC ≅ ΔDEF? Justify your answer.

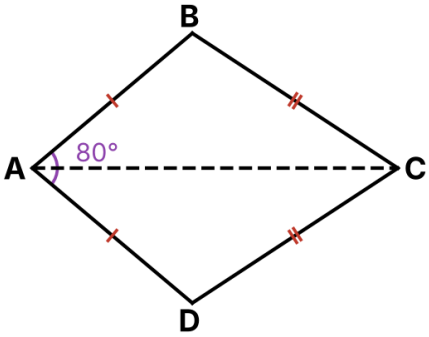


13. In quadrilateral ABCD, AB = AD, CB = CD and ∠BAD = 80°. AC is joined.

(a) Show that ΔABC ≅ ΔADC. Which criterion did you use?

(b) Find ∠BAC and ∠DAC.

(c) Is ∠ABC = ∠ADC? Give a reason.



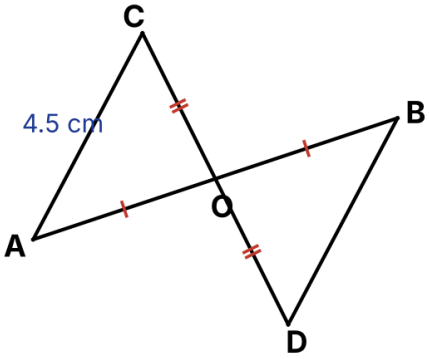
14. Line segments AB and CD cross each other at O such that OA = OB and OC = OD. AC = 4.5 cm.

(a) Which two angles at O are equal? Why?

(b) Show that ΔAOC ≅ ΔBOD. Name the criterion.

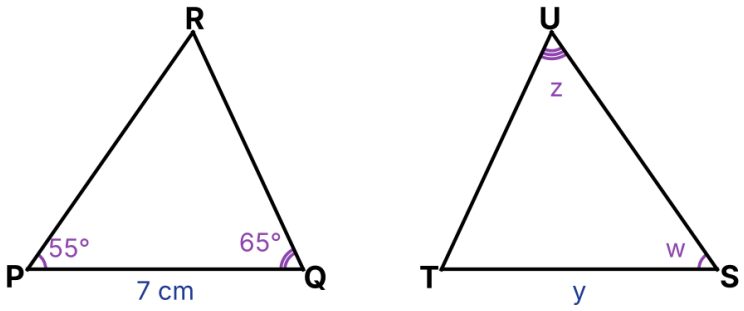
(c) Find the length of BD.

(d) Name the angle that is equal to ∠OAC.



Section C — Using Congruence to Find Unknown Parts

15. ΔPQR ≅ ΔSTU. Using the measurements in the figure, find y, w and z.

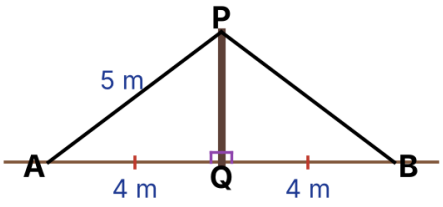


16. A vertical pole PQ stands on level ground. Two ropes PA and PB are tied from the top P to pegs A and B on the ground, on opposite sides of the pole, such that QA = QB = 4 m. Rope PA is 5 m long.

(a) Which criterion shows that ΔPQA ≅ ΔPQB? Write the three pairs of equal parts.

(b) Without measuring, find the length of rope PB.

(c) Is ∠PAQ = ∠PBQ? Give a reason.



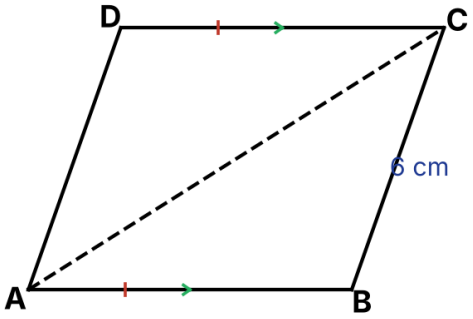
17. In quadrilateral ABCD, AB ∥ DC and AB = DC. Diagonal AC is drawn and BC = 6 cm.

(a) Which angle is equal to ∠BAC? Give a reason.

(b) Show that ΔABC ≅ ΔCDA. Name the criterion.

(c) Find the length of AD.

(d) What can you say about ∠ABC and ∠CDA?



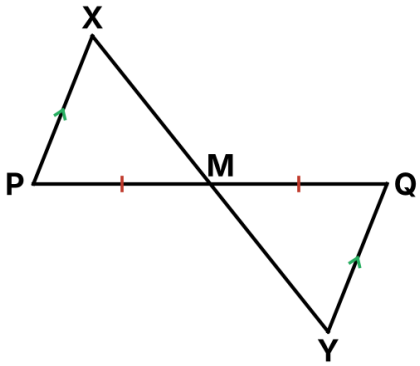
18. M is the midpoint of PQ. PX ∥ QY, and X, M, Y lie on a straight line.

(a) Name two pairs of equal angles in ΔPMX and ΔQMY and give reasons.

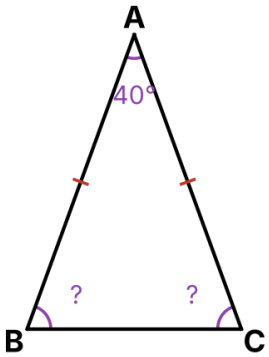
(b) Show that ΔPMX ≅ ΔQMY. Name the criterion.

(c) If XY = 10 cm, find MX.

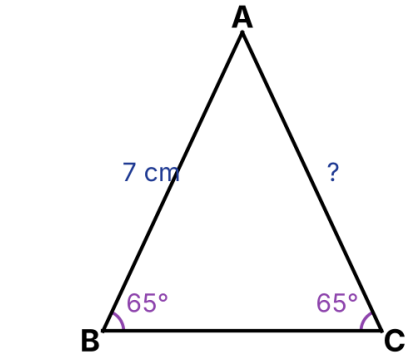
(d) If PX = 3.2 cm, find QY.



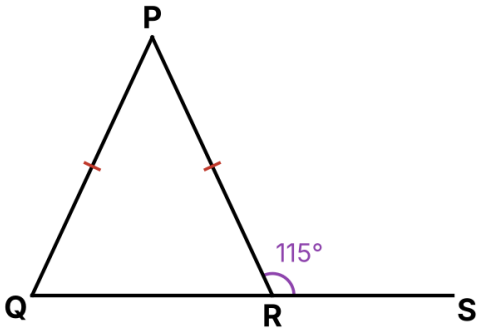
19. In ΔABC, AB = AC and ∠A = 40°. Find ∠B and ∠C.



20. In ΔABC, ∠B = ∠C = 65° and AB = 7 cm. Find the length of AC and the measure of ∠A.



21. In ΔPQR, PQ = PR. Side QR is extended to S, and ∠PRS = 115°. Find ∠PRQ, ∠PQR and ∠QPR.



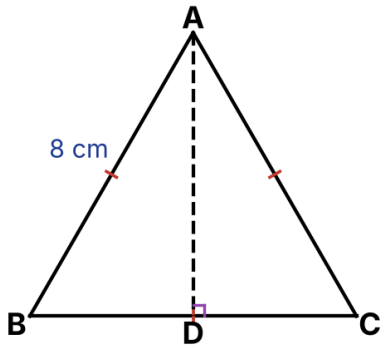
22. ABC is an equilateral triangle of side 8 cm. AD is perpendicular to BC.

(a) What is the measure of each angle of ΔABC? Why?

(b) Show that ΔABD ≅ ΔACD. Name the criterion.

(c) Find BD and DC.

(d) Find ∠BAD.



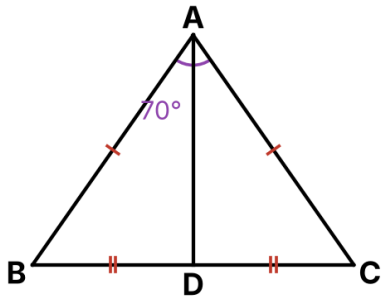
23. In ΔABC, AB = AC and D is the midpoint of BC. ∠BAC = 70°.

(a) Show that ΔABD ≅ ΔACD. Name the criterion.

(b) Find ∠BAD.

(c) Find ∠ADB. Explain your answer.

(d) Find ∠ABD.



Section D — Reasoning and Applications

24. Construct ΔABC with AB = 5 cm, BC = 6 cm and CA = 4 cm. Your friend constructs ΔPQR with PQ = 5 cm, QR = 6 cm and RP = 4 cm.

(a) If both triangles are cut out, will one fit exactly on the other? Which criterion supports your answer?

(b) Which angle of ΔPQR is equal to ∠B?

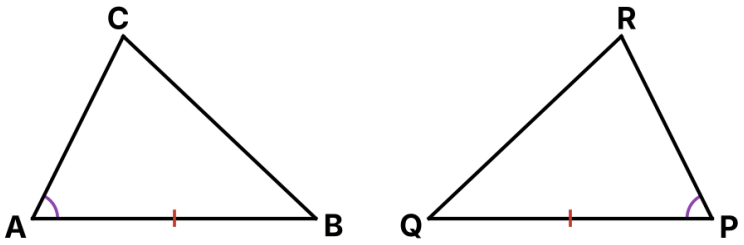
25. Meena says, “To be sure that two triangles are congruent, I must check that all six parts are equal.”

(a) Is she right? What is the least number of parts that need to be checked?

(b) Will *any* three equal parts do? Give one example of three equal parts that do **not** guarantee congruence.

26. In ΔABC and ΔPQR, AB = PQ and ∠A = ∠P (marked in the figure). What one more pair of equal parts is needed to prove ΔABC ≅ ΔPQR by

(a) the SAS criterion? (b) the ASA criterion? (c) the AAS criterion?

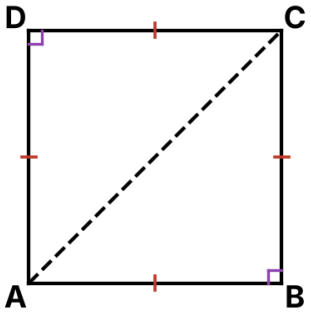


27. ABCD is a square and AC is a diagonal.

(a) Show that $\Delta ABC \cong \Delta ADC$. Name the criterion.

(b) What type of triangle is ΔABC (by its sides and by its angles)?

(c) Find $\angle BAC$ and $\angle BCA$.

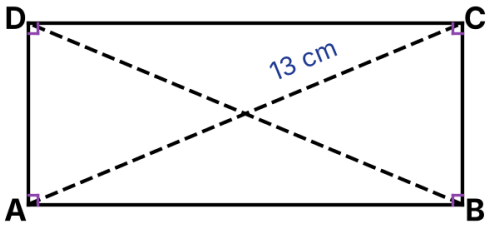


28. ABCD is a rectangle with diagonals AC and BD. AC = 13 cm.

(a) Show that $\Delta ABC \cong \Delta BAD$. Name the criterion.

(b) Find the length of BD.

(c) What property of a rectangle's diagonals does this show?

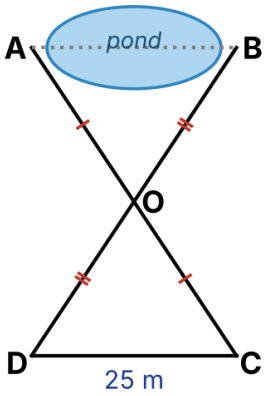


29. Arjun wants to know the distance AB across a pond, but he cannot walk across it. He fixes a point O on open ground. He walks from A through O to a point C such that OC = OA, and from B through O to a point D such that OD = OB. He measures CD = 25 m.

(a) Why is $\angle AOB = \angle COD$?

(b) Show that $\Delta AOB \cong \Delta COD$. Name the criterion.

(c) What is the distance AB across the pond?



30. (Challenge) ABC is an equilateral triangle. Points D, E and F lie on sides AB, BC and CA respectively, such that AD = BE = CF.

(a) Explain why DB = EC = FA.

(b) Show that ΔADF ≅ ΔBED. Name the criterion.

(c) Hence show that ΔDEF is also equilateral.

